

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
3.	(a)	(i)		3.92% of the population are heterozygotes = 3 marks If incorrect award 2 marks for 3.9 4 $2pq = 2 \times (0.02 \times 0.98)$ 0.0392 If incorrect award 1 mark for sight of $p = 1 - 0.02$ $p = 0.98$		3		3	3	3
		(ii)		{more resistant/ owtte} to malaria/ORA (1) Because parasite cannot reproduce/feed in (Tt individuals) RBC's (1) reject survive		2		2		
	(b)	(i)		{Undifferentiated/unspecialised/ multipotent} (can continue to divide) (1) Can {specialise and become / differentiate into} RBC's (1)		2		2		
		(ii)		Less/no chance of rejection/eq (1) Fewer/no ethical concerns (1)	2			2		
		(iii)		Any two (x1) from Identification of position of {Hb gene/ all genes} (1) Need to know base sequence of {normal/ replacement} gene (1) Need to know base sequence to RNA guide (1)		2		2		
				Question 3 total	2	9	0	11	3	3